CUSTOMER SALES ANALYSIS REPORT

Project Overview

The Customer Sales Analysis project was conducted to analyze customer purchasing behavior, product performance, regional sales trends, and overall business revenue. The objective of this analysis is to identify valuable customers, understand sales patterns, evaluate product performance, and provide actionable business recommendations for improving revenue and customer retention.

The analysis was performed using Python and Pandas, with data cleaning, aggregation, merging, pivot tables, and visualizations used to generate meaningful business insights.

Project Objectives

- Analyze customer purchasing patterns.
- Identify top-performing customers.
- Evaluate product performance.
- Analyze regional sales trends.
- Study monthly sales patterns.
- Create business intelligence visualizations.
- Generate actionable business recommendations.

Setup Instructions

Software Requirements

The following software and tools are required to run this project:

- Python 3.x
- Jupyter Notebook or Google Colab
- GitHub Account

Required Libraries

Install the required libraries using:

pip install pandas numpy matplotlib seaborn jupyter

Project Files

The following files must be present in the same directory:

- customer_analysis.ipynb
- sales_data.csv
- customer_data.csv
- analysis_report.pdf
- requirements.txt

Running the Project

Step 1

Download all project files.

Step 2

Open Jupyter Notebook or Google Colab.

Step 3

Load the notebook file:

customer_analysis.ipynb

Step 4

Run all notebook cells sequentially.

Step 5

Verify that:

- Datasets load successfully
- Data cleaning executes correctly
- Visualizations are generated
- Business insights are displayed

Code Structure

Project File Hierarchy

Customer-Sales-Analysis/

├── customer_analysis.ipynb

├── sales_data.csv

├── customer_data.csv

├── analysis_report.pdf

└── requirements.txt

Workflow Structure

1. Import Libraries
2. Load Datasets
3. Data Exploration
4. Data Cleaning
5. Data Merging
6. Aggregation Analysis
7. Customer Analysis
8. Product Analysis
9. Regional Analysis
10. Monthly Sales Analysis
11. Pivot Table Creation
12. Data Visualization
13. Business Insights
14. Recommendations

Technologies Used

Technology	Purpose
Python	Programming Language
Pandas	Data Manipulation

NumPy	Numerical Computation
Matplotlib	Data Visualization
Jupyter Notebook	Development Environment
GitHub	Version Control

Data Structures Used

Pandas DataFrame

Used for:

- Sales Dataset
- Customer Dataset
- Merged Dataset

Advantages:

- Efficient data storage
- Fast aggregation
- Easy filtering and transformation

Pandas Series

Used for:

- Product Revenue Analysis
- Customer Revenue Analysis
- Regional Revenue Analysis
- Monthly Sales Analysis

Algorithms and Techniques

Data Cleaning

Techniques Used:

- Duplicate removal
- Missing value handling
- Data validation

Data Merging

Method:

pd.merge()

Purpose:

Combine customer and sales datasets using customer identifiers.

Aggregation Functions

The following aggregations were performed:

- SUM()
- MEAN()
- COUNT()

Pivot Table Analysis

Method:

pd.pivot_table()

Purpose:

Summarize sales revenue across products and regions.

Visualization Techniques

- Bar Chart
- Line Chart
- Pie Chart
- Histogram

System Architecture

Sales Dataset
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▼
Data Cleaning
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▼

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Customer Dataset
│
▼
Data Merge
│
▼
Data Analysis
┌─────┼─────┐
▼ ▼ ▼
Customer Product Region
Analysis Analysis Analysis
│
▼
Pivot Table
│
▼
Visualizations
│
▼
Business Insights
│
▼
Recommendations
```

Executive Summary

This project analyzed 100 sales transactions and customer information records to evaluate sales performance and customer behavior.

Key Performance Indicators (KPIs)

Metric	Value
Total Revenue	12,365,048
Total Transactions	100
Total Customers	100
Average Order Value	123,650.48

Top Customer	CUST016
Top Customer Revenue	373,932
Best Selling Product	Laptop
Highest Sales Region	North
Highest Sales Month	March 2024

The analysis revealed that laptop sales generated the highest revenue among all products, while the North region contributed the largest share of total sales. Customer CUST016 was identified as the highest-value customer.

Dataset Description

The project uses two datasets:

Sales Dataset

The sales dataset contains transaction-level information including:

- Date
- Product
- Quantity
- Price
- Customer_ID
- Region
- Total_Sales

Total Records: 100

Customer Dataset

The customer dataset contains customer-related information including:

- CustomerID
- Tenure

- MonthlyCharges
- TotalCharges
- Contract
- PaymentMethod
- PaperlessBilling
- SeniorCitizen
- Churn

Total Records: 500

Methodology

The project followed a complete data analysis pipeline consisting of:

Step 1: Data Loading

Both datasets were loaded using Pandas.

Step 2: Data Cleaning

The following cleaning operations were performed:

- Removed duplicate records
- Checked for missing values
- Filled missing values where necessary
- Verified dataset integrity

Step 3: Data Merging

The sales and customer datasets were merged using customer identifiers to create a unified dataset for analysis.

Step 4: Data Aggregation

Three major aggregation techniques were used:

Sum Aggregation

Used to calculate:

- Total Revenue
- Product Revenue
- Regional Revenue

Mean Aggregation

Used to calculate:

- Average Order Value

Count Aggregation

Used to calculate:

- Total Transactions
- Total Customers

Step 5: Pivot Table Analysis

Pivot tables were used to summarize revenue by:

- Region
- Product

This allowed comparison of product performance across different regions.

Customer Analysis

Top Customers

The top customers were identified based on total revenue contribution.

Top 10 Customers

1. CUST016 – 373,932
2. CUST007 – 363,870
3. CUST083 – 350,888
4. CUST073 – 349,510
5. CUST020 – 333,992
6. CUST084 – 324,144

7. CUST070 – 318,762
8. CUST005 – 318,680
9. CUST065 – 312,564
10. CUST028 – 304,465

Observation

A small group of customers contributes a significant portion of total revenue. These customers should be targeted through loyalty programs and personalized marketing campaigns.

Product Performance Analysis

Revenue by product category:

Product	Revenue
Laptop	3,889,210
Tablet	2,884,340
Phone	2,859,394
Headphones	1,384,033
Monitor	1,348,071

Observation

Laptops generated the highest revenue and represent the strongest-performing product category.

Business Recommendation

- Increase marketing for laptops.
- Bundle laptops with accessories.
- Introduce premium laptop models.

Regional Sales Analysis

Revenue generated by region:

Region	Revenue
North	3,983,635
South	3,737,852
East	2,519,639
West	2,123,922

Observation

The North region contributed the highest revenue.

Business Recommendation

- Expand promotional activities in North.
- Replicate successful strategies from North in other regions.

Monthly Sales Trend Analysis

Monthly sales performance:

Month	Revenue
January 2024	4,120,524
February 2024	2,656,050
March 2024	4,485,006
April 2024	1,103,468

Observation

March recorded the highest sales, indicating strong seasonal demand during this period.

Business Recommendation

- Increase inventory before high-demand periods.

- Schedule marketing campaigns before peak months.

Pivot Table Analysis

The pivot table summarized sales revenue by region and product.

Key findings:

- North region generated exceptional laptop sales.
- South region performed strongly in phone sales.
- East region had balanced sales across products.
- West region showed moderate performance in all categories.

Visualizations Created

The project includes the following visualizations:

1. Bar Chart

Top 10 Customers by Revenue

Purpose:

- Identify high-value customers

2. Line Chart

Monthly Sales Trend

Purpose:

- Analyze sales growth over time

3. Pie Chart

Regional Sales Distribution

Purpose:

- Understand market contribution by region

4. Histogram

Sales Distribution

Purpose:

- Analyze transaction distribution and variability

Key Findings

1. Total revenue generated was 12.36 million.
2. Customer CUST016 was the highest-value customer.
3. Laptops were the best-selling product category.
4. North region generated the highest revenue.
5. March 2024 recorded the highest monthly sales.
6. Revenue is concentrated among a relatively small group of customers.
7. Regional performance varies significantly across markets.

Business Recommendations

Customer Retention

- Implement customer loyalty programs.
- Reward repeat customers.
- Offer personalized discounts.

Product Strategy

- Focus on high-performing laptop products.
- Introduce bundled product packages.
- Optimize pricing strategies.

Regional Growth

- Increase marketing investment in North and South regions.
- Develop targeted campaigns for East and West regions.

Sales Planning

- Prepare inventory for seasonal demand.
- Use monthly trend analysis for forecasting.

Data Strategy

- Continue monitoring customer behavior.
- Build customer segmentation models.
- Use predictive analytics for future growth.

Conclusion

The Customer Sales Analysis project successfully identified key sales trends, top-performing customers, regional opportunities, and product performance patterns. The analysis demonstrates how data-driven decision making can improve customer retention, optimize product strategy, and increase overall business profitability.

The findings from this project provide valuable insights that can support strategic planning and future business growth.

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CUSTOMER SALES ANALYSIS PROJECT
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Datasets Loaded Successfully

Sales Shape: (100, 7)
Customer Shape: (500, 9)

Sales Columns
Index(['Date', 'Product', 'Quantity', 'Price', 'Customer_ID', 'Region',
       'Total_Sales'],
      dtype='object')

Customer Columns
Index(['CustomerID', 'Tenure', 'MonthlyCharges', 'TotalCharges', 'Contract',
       'PaymentMethod', 'PaperlessBilling', 'SeniorCitizen', 'Churn'],
      dtype='object')

Missing Values in Sales
Date           0
Product        0
Quantity       0
Price          0
Customer_ID    0
Region         0
Total_Sales    0
dtype: int64

Missing Values in Customers
CustomerID          0
Tenure              0
MonthlyCharges      0
TotalCharges        0
Contract            0
PaymentMethod       0
PaperlessBilling    0
SeniorCitizen       0
Churn               0
dtype: int64

Data Cleaning Completed
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Merged Dataset Shape: (100, 16)

SUMMARY METRICS
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Total Revenue: 12365048
Average Order Value: 123650.48
Total Customers: 100
Total Transactions: 100

TOP 10 CUSTOMERS
Customer_ID
CUST016    373932
CUST007    363870
CUST083    350888
CUST073    349510
CUST020    333992
CUST084    324144
CUST070    318762
CUST005    318680
CUST065    312564
CUST028    304465
Name: Total_Sales, dtype: int64

PRODUCT SALES
Product
Laptop        3889210
Tablet        2884340
Phone         2859394
Headphones    1384033
Monitor       1348071
Name: Total_Sales, dtype: int64

REGIONAL SALES
Region
North    3983635
South    3737852
East     2519639
West     2123922
Name: Total_Sales, dtype: int64
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MONTHLY SALES
Month
2024-01    4120524
2024-02    2656050
2024-03    4485006
2024-04    1103468
Name: Total_Sales, dtype: int64

PIVOT TABLE
Product  Headphones   Laptop  Monitor    Phone   Tablet
Region
East         288361   221946   642870   506828   859634
North        107091  1798206   397100   489284  1191954
South        512168  1373120    39924  1471428   341212
West         476413   495938   268177   391854   491540
```
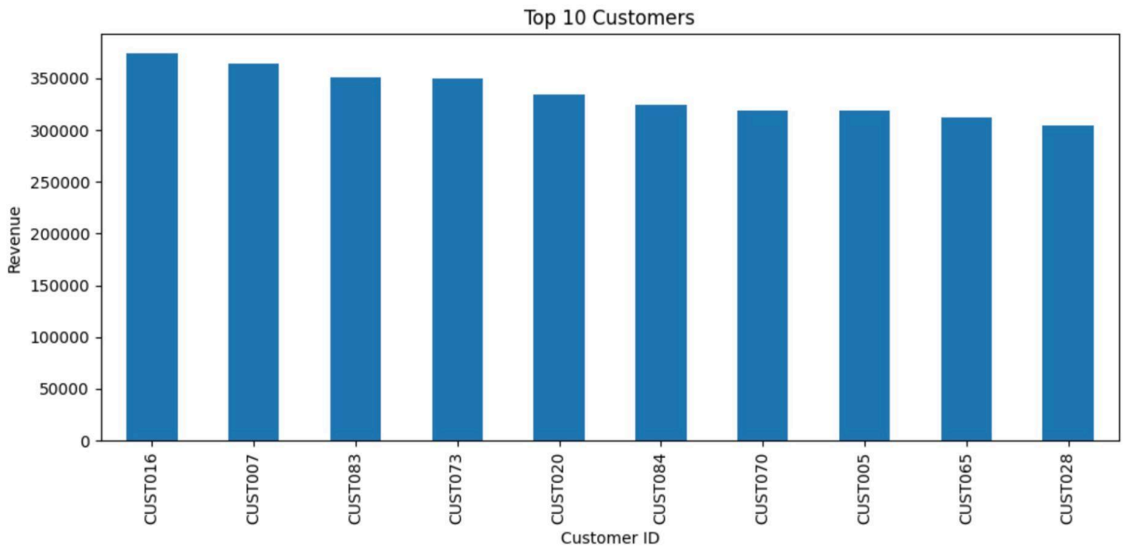
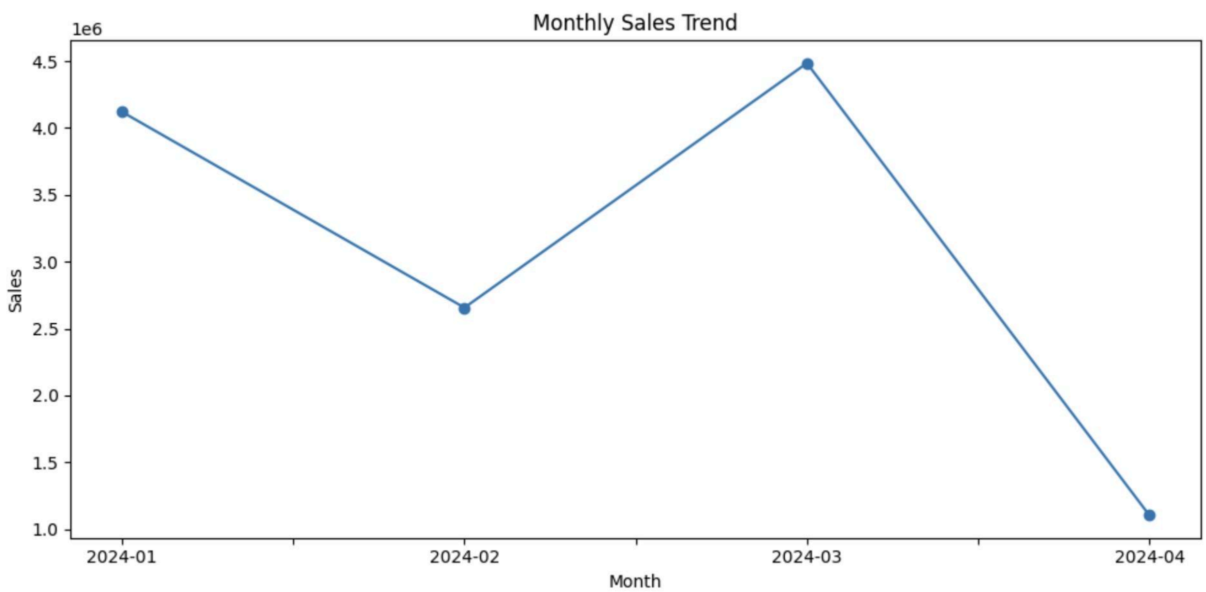
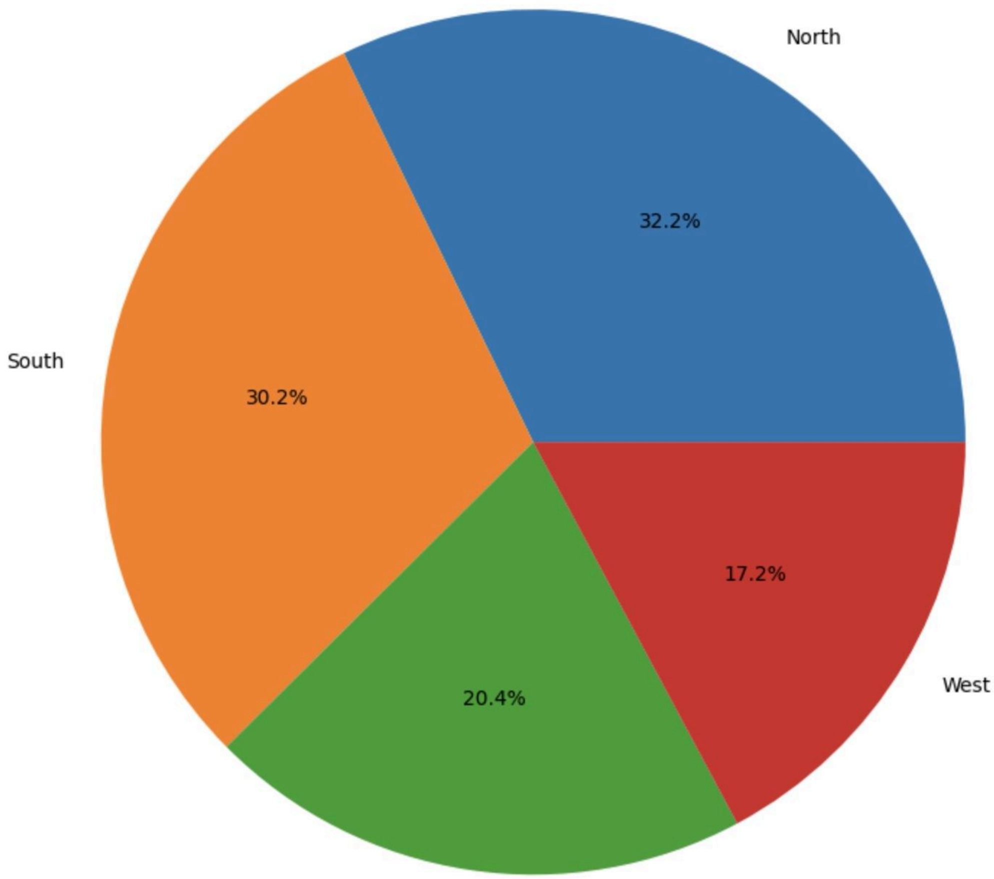
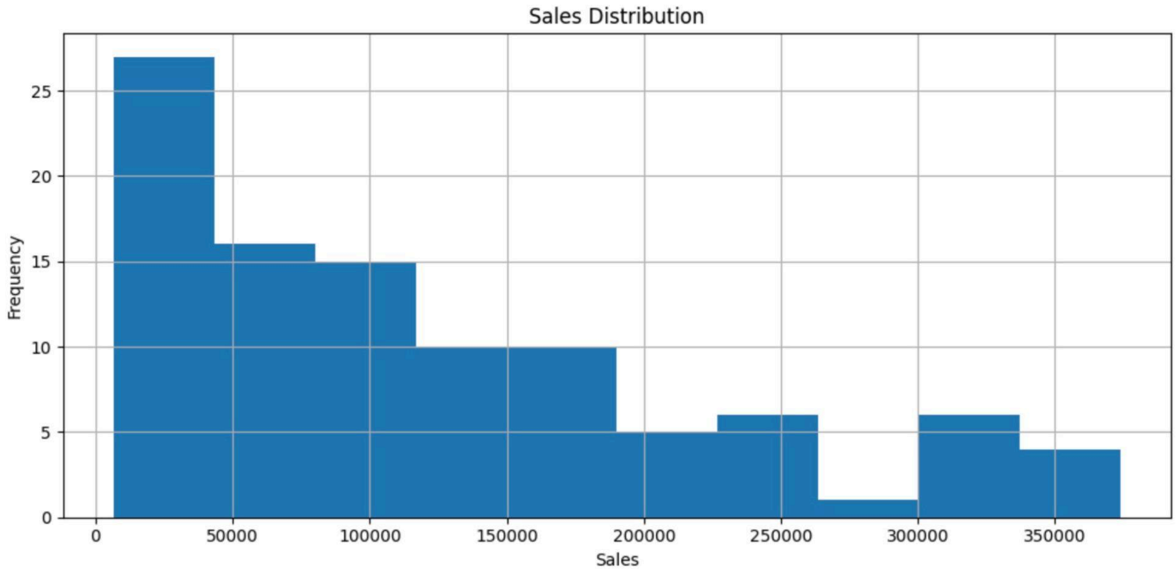


Top 10 Customers
Revenue
350000
300000
250000
200000
150000
100000
50000
0
CUST016
CUST007
CUST083
CUST073
CUST020
CUST084
CUST070
CUST005
CUST065
CUST028
Customer ID


Monthly Sales Trend
1e6
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
Sales
2024-01
2024-02
2024-03
2024-04
Month

Regional Sales Distribution



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BUSINESS INSIGHTS
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Top Customer: CUST016
Revenue Generated: 373932

Best Selling Product:
Laptop

Highest Sales Region:
North

Recommendations
1. Reward high-value customers.
2. Focus marketing in North region.
3. Promote laptops aggressively.
4. Monitor monthly sales trends.
5. Use customer segmentation for future campaigns.

PROJECT COMPLETED SUCCESSFULLY
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